MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8281

Ⓑ This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Vertical, base down or up
Weight: 12.8 pounds (5.8 kg) nominal

ABSOLUTE RATINGS: F = 110 MHz

Parameter:	Ef	Eb	Ec2	Ec1	Ib	Pg1	Pg2	Pp	Anode core & seal T	Cooling
Unit:	V ac	kV dc	kV dc	kV dc	A dc	W	W	kW	°C	---
Maximum:										(Note 1)
C Teleg:	6.3 ±5%	10	2	-1.5	5	200	450	15	250	---
C Telep: (anode mod)	6.3 ±5%	8	1.5	-1.5	4	200	450	10	250	---
Class AB:	6.3 ±5%	10	2	---	6	200	450	15	250	---
TEST CONDITIONS:	6.3	2	0.75	Adj	1	---	---	---	---	Note 2

Ⓑ GENERAL:

Qualification - Required

Ⓑ denotes changes

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Filament current	-	t = 120 ±15	0.65	II	If	152	168	A ac
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-110	-146	V dc
1266	Total grid current	-		0.65	II	Ic1	---	-25	µA dc
1256	Electrode current (screen)	-		0.65	II	Ic2	---	25	mA dc
1231	Peak emission	-	eb = ec2 = ec1 = 2.5 kv	0.65	II	is	90	---	a
1266	Primary grid emission (grid)	-	Pg1 = 200 W; t = 120 max or until stable; anode and g2 floating	0.65	II	Isg1	---	-500	µA dc
1266	Primary grid emission (screen)	-	Pg2 = 450 W; Ec1 = 0 V dc; t = 120 max or until stable; anode floating	0.65	II	Isg2	---	-500	µA dc
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance (ground cathode connection)	-		---	---	Cin	154	167	pF
						Cout	22	27	pF
						Cgp	---	2	pF
1331	Direct-interelectrode capacitance (ground grid connection)	-		---	---	Cin	62	72	pF
						Cout	23	28	pF
						Cpk	---	0.30	pF
1372	Current division (method B, short pulse)	-	Eb = Ec2 = 2,000 V dc; Ec1 = -800 V dc; egk/ib = 19 a	---	---	egk	---	0	v
				---	---	ic2	---	3.2	a
---	Power output	6	Class AB1 amp; F = 1 MHz (min); Eb = 9 kV dc; Ec2 = 2 kV dc; Ec1/Ibo = 0.1 A dc; Eg1/Ib = 3.7 A dc; RL = 1,125 ±5%; anode tank Q = 10 to 15	---	---	Po	20	---	kW (useful)

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 3								
---	Service-life guarantee	3		---	---	---	---	---	---
1042	Shock, specified pulse	4	No voltages applied; shock = 11 ms half-sine; accel = 15 G peak (min); impacts = 6 (3 each X and Z axes)	---	---	---	---	---	---
1032	Vibration, mechanical	4	No voltages applied; accel = 2 G peak (min); F = 10 to 50 Hz, ascending only; sweep t = 3 to 8 minutes; 1 sweep each X and Y axes	---	---	---	---	---	---
---	Shock and vibration, mechanical end points:								
1261	Electrode voltage (grid)	-		---	---	Ec1	-100	-146	V dc
1266	Total grid current	-		---	---	Ic1	---	-30	µA dc
1301	Filament current	5		---	---	ΔIf	---	3	A ac

NOTES:

1. Minimum airflow requirements for incoming air at 50°C maximum at sea level, for operation under 30 MHz, are shown. Additional cooling may be required for operation above 30 MHz. In all cases of operation a socket which provides for forced-air cooling of the base must be used, such as the EIMAC SK-300A, or equivalent, used with the EIMAC SK-316 Air Chimney, or equivalent, with air flowing in a base-to-anode direction. Where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial. Cooling air should be applied before or simultaneously with the application of electrode voltages, including the filament, and should normally be maintained for a short period after all voltages are removed to allow for tube cool-down. The cooling data shown is for the tube in a SK-300A socket with a SK-316 Air Chimney.

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Anode dissipation	Sea level		10,000 feet	
	Airflow (cfm)	Approximate pressure drop ($In.H_2O$)	Airflow (cfm)	Approximate pressure drop ($In.H_2O$)
7,500 W	220	0.4	320	0.6
12,500 W	555	2.5	810	3.6
15,000 W	775	5.0	1,130	7.3

2. In all electrical tests involving application of filament voltage an air-system socket and chimney may be used and forced-air cooling is allowable.

3. The tube manufacturer warrants the tube for 1 year from date of shipment, or 1,000 hours of filament life, whichever first elapses. This warranty applies only when the tube is operated within the maximum ratings (see "Absolute Ratings" of MIL-E-1). A defective tube shall either be replaced, or at the option of the manufacturer, a credit shall be made in the amount of the original purchase price pro rated on the basis of 1,000 hours of "filament-on" time.

4. Testing shall be performed every 6 months, with sampling as follows:

$n_1 = 4$ $c_1 = 0$

$n_2 = 4$ $c_2 = 1$;

where c_2 represents the total allowable failures for the first and second samples combined.

Separate samples may be used at the option of the manufacturer. None of the listed tests shall be considered destructive except in case of failure. In the event of failure after double sampling, that specific test shall become quality conformance inspection, part 2; after three consecutive successful submissions, the testing may revert to the quality conformance inspection, part 3 tests.

5. Any change in filament current resulting from the vibration or shock testing (considered individually) shall not exceed the specified limit for ΔIf.

6. During this test the tube shall be operated as a Class AB1 amplifier; the control grid shall not be driven positive, as indicated by grid current flow.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Navy - AS, OS, MC, CG
Air Force - 11, 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3331)

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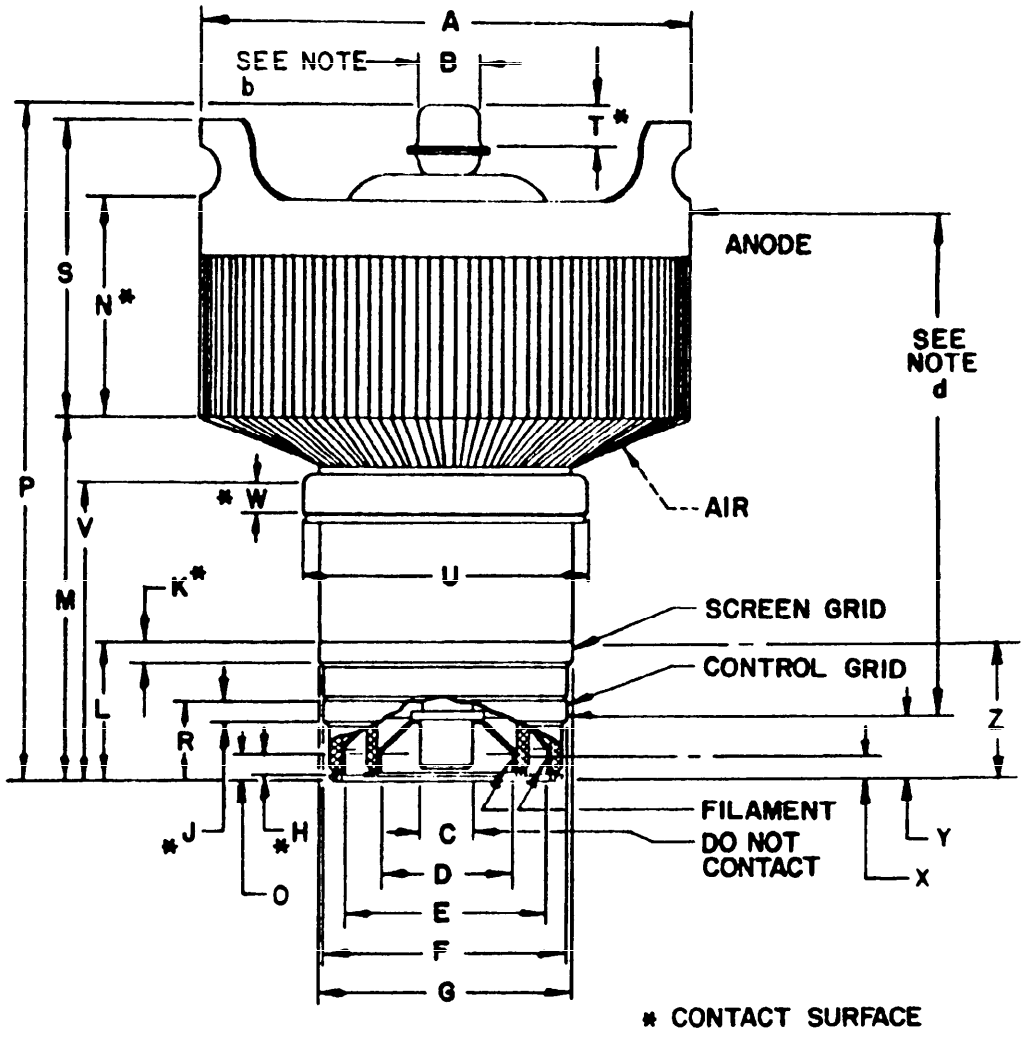


FIGURE 1. Outline drawing of electron tube type 8281.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
C	.600 (15.24)	.760 (19.30)
D	1.896 (48.16)	1.936 (49.17)
E	3.133 (79.58)	3.173 (80.59)
F	3.792 (96.32)	3.832 (97.33)
G	3.980 (101.09)	4.020 (102.11)
H	.188 (4.78)	--- ---
J	.188 (4.78)	--- ---
K	.188 (4.78)	--- ---
L	1.695 (43.05) BASIC (See note e)	
O	.355 (9.02) BASIC (See note e)	
P	9.000 (228.60)	9.375 (238.12)
R	.920 (23.37) BASIC (See note e)	
T	.375 (9.52)	--- ---
U	4.406 (111.91)	4.468 (113.49)
V	3.718 (94.44)	3.781 (96.04)
W	.219 (5.56)	--- ---
Quality conformance inspection, part 3 (See note c)		
A	7.460 (189.48)	7.580 (192.53)
B	.855 (21.72)	.895 (22.73)
M	4.550 (115.57)	4.783 (121.49)
N	2.412 (61.26)	2.788 (70.82)
S	3.560 (90.42)	3.684 (93.57)
Reference dimensions (See notes f & g)		
X	.260 (6.60)	
Y	.826 (20.98)	
Z	1.600 (40.64)	

FIGURE 1. Outline drawing of electron tubes type 8281 - Continued.

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NOTES:

a. The total indicator reading (T.I.R.) (the sum of the positive and negative deflection shown by the indicator when measuring the eccentricity of the surface with respect to another, with the reference axis established) of the screen grid and filament contact surfaces shall not exceed .040 (1.02 mm) with respect to the control grid and anode contact surfaces when the latter surfaces are rotated on rollers at the points indicated by the arrows. Quality conformance inspection part 2, shall apply.

b. Top cap outline optional provided it meets requirements of dimensions B and T.

c. Dimensions shall be checked every 6 months, with sampling as follows:

$n_1 = 4$ $c_1 = 0$ where c_2 represents the total allowable
$n_2 = 4$ $c_2 = 1$; failures for the first and second samples combined.

Separate samples may be used at the option of the manufacturer. None of the listed tests shall be considered destructive except in case of failure. In the event of failure after double sampling, that specific test shall become quality conformance inspection, part 2; after three consecutive successful submissions, the testing may revert to the quality conformance inspection, part 3 tests.

d. The T.I.R. of the screen grid and filament contact surfaces shall not exceed .040 (1.02 mm) with respect to the control grid and anode contact surface when the latter surfaces are rotated on rollers at points indicated by the arrows.

e. Basic dimension is a numerical value used to describe the theoretically exact size, shape or location of a feature or datum target. It is the basis from which permissible variations are established by tolerances on other dimensions, in notes or by feature control symbols.

f. Optimum filament and grid connector heights for socket design purposes.

g. Reference or nominal dimensions are listed for information only, and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 8281 - Continued.

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